

TASK 3: Artificial Intelligence Chatbot

Java Programming — CodeAlpha Internship

Task Requirements

- Create a Java-based chatbot for interactive communication
- Use Natural Language Processing (NLP) techniques
- Implement machine learning logic or rule-based answers
- Train the bot to respond to frequently asked questions
- Integrate with a GUI or web interface for real-time interaction

Design Approach

- Rule-based NLP: HashMap stores keyword-to-response pairs
- Exact match first, then partial keyword scanning for flexibility
- Fallback response if no keyword found
- Conversation history maintained in ArrayList
- Covers Java, OOP concepts, greetings, and general chat

Complete Java Source Code

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.Scanner;

// Represents a Q&A pair
class QAPair {
    String question;
    String answer;

    QAPair(String question, String answer) {
        this.question = question;
        this.answer = answer;
    }
}

public class AIChatbot {

    // Knowledge base using HashMap for rule-based matching
    private HashMap<String, String> knowledgeBase = new HashMap<>();
    private ArrayList<String> conversationHistory = new ArrayList<>();

    public AIChatbot() {
        // Greetings
        knowledgeBase.put("hello", "Hello! I'm CodeAlpha AI. How can I assist you today?");
        knowledgeBase.put("hi", "Hi there! What can I help you with?");
        knowledgeBase.put("hey", "Hey! Great to meet you. How can I help?");

        // General questions
        knowledgeBase.put("what is your name", "I'm CodeAlpha AI Chatbot, your Java-based assistant!");
    }
}
```

```

        knowledgeBase.put("who are you", "I'm an AI chatbot built with Java as part of
the CodeAlpha internship.");
        knowledgeBase.put("how are you", "I'm running perfectly! Thanks for asking. How
about you?");

        // Java related
        knowledgeBase.put("what is java", "Java is a high-level, object-oriented
programming language developed by Sun Microsystems in 1995.");
        knowledgeBase.put("what is oop", "OOP stands for Object-Oriented Programming. It
uses objects and classes to structure code.");
        knowledgeBase.put("what is inheritance", "Inheritance allows a class to inherit
properties and methods from a parent class using 'extends'.");
        knowledgeBase.put("what is polymorphism", "Polymorphism allows objects of
different types to be treated as the same type through interfaces or parent classes.");
        knowledgeBase.put("what is encapsulation", "Encapsulation is wrapping data and
methods in a single unit (class) and restricting direct access using private fields.");

        // Farewells
        knowledgeBase.put("bye", "Goodbye! Have a great day!");
        knowledgeBase.put("exit", "Thank you for chatting. See you next time!");
        knowledgeBase.put("goodbye", "Goodbye! Stay curious and keep coding!");

        // Fallback topics
        knowledgeBase.put("help", "You can ask me about Java, OOP concepts, or just chat!
Try: 'what is java', 'hello', etc.");
        knowledgeBase.put("thank you", "You're welcome! I'm happy to help.");
        knowledgeBase.put("thanks", "No problem! Let me know if you need anything
else.");
    }

    // NLP-style matching: keyword scan
    public String getResponse(String input) {
        String lower = input.toLowerCase().trim();

        // Exact match first
        if (knowledgeBase.containsKey(lower)) {
            return knowledgeBase.get(lower);
        }

        // Keyword-based partial matching
        for (String key : knowledgeBase.keySet()) {
            if (lower.contains(key)) {
                return knowledgeBase.get(key);
            }
        }

        // Fallback response
        return "I'm not sure about that. Try asking about Java, OOP, or type 'help!'";
    }

    public void chat() {
        Scanner sc = new Scanner(System.in);
        System.out.println("\n=== CodeAlpha AI Chatbot ===");
        System.out.println("Type 'bye' or 'exit' to quit.\n");

        while (true) {
            System.out.print("You: ");
            String input = sc.nextLine();

```

```

        conversationHistory.add("You: " + input);

        if (input.equalsIgnoreCase("exit") || input.equalsIgnoreCase("bye")
            || input.equalsIgnoreCase("goodbye")) {
            String farewell = getResponse(input);
            System.out.println("Bot: " + farewell);
            break;
        }

        String response = getResponse(input);
        System.out.println("Bot: " + response);
        conversationHistory.add("Bot: " + response);
    }
    sc.close();
}

public void printHistory() {
    System.out.println("\n--- Conversation History ---");
    for (String line : conversationHistory)
        System.out.println(line);
}

public static void main(String[] args) {
    AIChatbot bot = new AIChatbot();
    bot.chat();
    bot.printHistory();
}
}

```

Sample Conversation

```

=== CodeAlpha AI Chatbot ===
Type 'bye' or 'exit' to quit.

```

```

You: hello
Bot: Hello! I'm CodeAlpha AI. How can I assist you today?
You: what is java
Bot: Java is a high-level, object-oriented programming language...
You: what is inheritance
Bot: Inheritance allows a class to inherit properties...
You: bye
Bot: Goodbye! Have a great day!

```